

**Course Breakdown for
Physics Bachelors Degree Programme
For In-Service Trainees**

Based on Harmonized Curriculum for Physics B.Sc Degree

*(Revised mode of delivery according to the need of physics
department)*

General Introduction

This is the proposed course breakdown for summer in-service physics degree programme for teachers. During the preparation of the course breakdown the quality issue is given much attention and the fact that the trainees being teachers having diploma in physics or related subjects is considered, hence they are exempted from some courses. Moreover they are exempted from some advanced elective courses considering that they are already engaged in the teaching profession. Even though three courses of contemporary applications having much relevance to their teaching in schools are selected thoroughly. In the programme a total of 102 credit hours subject areas (major areas + supportive courses) and common courses are to be given. Moreover, there are 33 credit hours of professional courses to be given in the programme. Out of the total of 135 credits, 89 credits are to be given face to face in six summers and the remaining 46 credit hours are to be given in 5 intermediate winters.

List of Required Courses

Course N.	Course Title	Credits	Course Type
Phys 201	Mechanics	4	Major Area
Phys 202	Electromagnetism	4	Major Area
Phys 211	Experimental Physics I	2	Major Area
Phys 212	Experimental Physics II	2	Major Area
Phys 242	Modern Physics	3	Major Area
Phys 301	Mathematical Methods of Physics I	3	Major Area
Phys 302	Mathematical Methods of Physics II	3	Major Area
Phys 312	Experimental Physics III	2	Major Area
Phys 321	Statistical Physics I	3	Major Area
Phys 331	Classical Mechanics I	3	Major Area
Phys 342	Quantum Mechanics I	3	Major Area
Phys 353	Electronics I	3	Major Area
Phys 203	Waves & Optics	2	Major Area
Phys 371	Modern Optics	3	Major Area
Phys 376	Electrodynamics I	3	Major Area
Phys 382	Nuclear Physics I	3	Major Area
Phys 402	Introduction to Computational Physics	3	Major Area
Phys 411	Experimental Physics IV	2	Major Area
Phys 422	Statistical Physics II	3	Major Area
Phys 432	Classical Mechanics II	3	Major Area
Phys 441	Quantum Mechanics II	3	Major Area
Phys 451	Solid State Physics I	3	Major Area
Phys ----	Elective Courses (I, II, III)	9	Major Area
Phys 461	Sustainable Sources of Energy	2	Major Area
Phys 476	Electrodynamics II	3	Major Area
Phys 492	Research Methods & Senior Project	3	Major Area
Math 261	Calculus I	4	Supportive
Phys 262	Calculus II	4	Supportive
Math 325	Linear Algebra	3	Supportive
Comp 201	Introduction to Computer Applications	3	Supportive
Comp 271	Introduction to Programming	4	Supportive
Enla 201	Communicative English Skills	3	Common Course
Enla 202	Writing Skills	3	Common Course
CvEt 202	Civics & Ethics	3	Common Course
	School & Society	3	Professional
	Psychological Foundations of Learning & Development	3	Professional
	Secondary School Curriculum & Instruction	4	Professional
	Assessment & Evaluation of Learning	3	Professional
	Subject Area Methods	3	Professional
	Teaching in Multicultural Setting	2	Professional
	English for Secondary School Teaching & Learning	3	Professional
	Action Research Project	2	Professional
	Teachers as Reflective Practitioners	3	Professional
	Subject Area Methods II	2	Professional
	Special Needs Education	3	Professional
	Instructional Technology	2	Professional
Total Credits		0	

Total Summary of Graduation Requirements

S.N	Course Category	Mode of Delivery		Total
		Summer(Face to Face)	Winter (Distance)	
1	Major Areas	49	31	80
2	Supportive (Maths + Computer)	15	3	18
3	Common Courses	3	6	9
4	Professional Courses	22	11	33
Overall Total		0	89	51

Course Breakdown into 6 summers & 5 winters

Kiremt I (face to face)

Course No	Course title	Credits	Lecture	Lab	Tutorial
Phys 201	Mechanics	4	4		2
Phys 211	Experimental Physics I	2		3	1
Math 261	Calculus I	4	4		2
Comp 201	Introduction to Computer Applications	3	3	1	
Enla 201	Communicative English skills	3	3		
Total		0	14	4	4

Bega I (Distance)

Course No	Course title	Credits	Lecture	Tutorial
CVet 202	Civics & Ethics	3	3	
Phys 203	Waves & Optics	2	2	1
Elna 202	Writing skills	3	3	
Total		8	8	1

Kiremt II (Face to Face)

Course No	Course title	Credits	Lecture	Lab	Tutorial
Phys 202	Electromagnetism	4	4	-	2
Comp 271	Introduction to Programming	4	4	1	-
Math 262	Calculus II	4	4	-	2
Phys 331	Classical Mechanics I	3	3	-	1
Phys 212	Experimental Physics II	2		3	-
Total		17	17	4	5

Bega II (Distance)

Course No	Course title	Credits	Lecture	Tutorial
Math 325	Linear Algebra	3	3	
Phys ----	Elective I	3	3	
Phys 242	Introduction to Modern physics	3	3	1
Total		9	9	1

Kiremt III (Face to Face)

Course No	Course title	Credits	Lecture	Lab	Tutorial
Phys 321	Statistical physics I	3	3	-	1
Phys 371	Modern Optics	3	3	-	1
Phys 301	Mathematical Methods of Phys I	3	3	-	1
Phys 342	Quantum Mechanics I	3	3	-	1
Phys 451	Solid State Physics I	3	3	-	1
Total		15	15		5

Bega III (Distance)

Course No	Course title	Credits	Lecture	Tutorial
Phys 461	Sustainable sources of Energy	2	2	
Phys ---	Elective II	3	3	
Total		5	5	

Kiremt IV (Face to Face)

Course No	Course title	Credits	Lecture	Lab	Tutorial
Phys 382	Nuclear Physics I	3	3		1
Phys 312	Experimental Physics III	2		3	1
Phys 353	Electronics I	3	3		1
Phys 376	Electrodynamics I	3	3		1
Phys 302	Mathematical Methods of Phys II	3	3		1
Total		0	12	3	5

Bega IV (Distance)

Course No	Course title	Credits	Lecture	Tutorial
Phys 492	Research Methods & Senior Project	3	3	
Phys ---	Elective III	3	3	
Total		6	6	

Kiremt V (Face to Face)

Course No	Course title	Credits	Lecture	Lab	Tutorial
Phys 441	Quantum Mechanics II	3	3		1
Phys 411	Experimental Physics IV	2		3	
Phys 402	Introduction to Computational Phys.	3	3		1
Phys 422	Statistical Physics II	3	3		
Phys 476	Electrodynamics II	3	3		1`
	Teachers as Reflective Practitioners	3	3		
Total		17	15	3	3

Bega IV (Distance)

Course No	Course title	Credits	Lecture	Tutorial
	Subject Area Methods I	3	3	
	Secondary School Curriculum & Instruction	4	4	
	School & Society	3	3	
	Psychological Foundations of Learning & Development	3	3	
	Action Research Project	2	2	
Total		15	15	

Kiremt VI (Face to Face)

Course No	Course title	Credits	Lecture	Lab	Tutorial
	Teaching in Multicultural Settings	2	2		
	English for Secondary School T & L	3	3		
	Special Needs Education	3	3		
	Instructional Technology	2	2		
	Subject Area Methods II	2	2		
	Assessment and Evaluation	3	3		
Total		15	15	15	

Elective courses

	Course code	Course title	Credits	lectures	lab	Tutorial
Elective I	Phys 316	Metrology I	3	3		1
	Phys 367	Environmental Physics	3	3		1
	Phys 369	General geophysics	3	3		1
Elective II	Phys 463	Atmospheric physics	3			1
	Phys 471	Introduction to Laser physics	3	3		1
	Phys 415	Metrology II	3	3		1
	Phys 468	Exploration Geophysics	3	3		1
Elective III	Phys 437	Astronomy	3	3		1
	Phys 439	Space physics	3			1
	Phys 452	Solid state physics II	3	3		1
	Phys 484	Radiation physics	3	3		1
	Phys 384	Introduction to Medical Physics	3	3		1
	Phys 416	Metrology III	3	3		1